

Lecture 27

Particle Swarm Optimisation

COMP532 – Machine Learning and Bioinspired Optimisation

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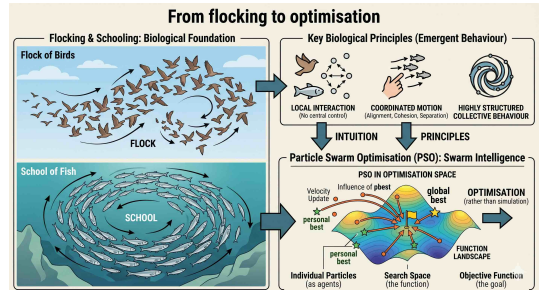


Lecture goals

- Introduce **Particle Swarm Optimisation (PSO)** as a population-based method for continuous search.
- Explain how **self-memory** and **social influence** jointly guide optimisation.
- Understand the canonical PSO update rule and its design trade-offs.
- Connect PSO to the broader theme of **distributed search without gradients**.

From flocking to optimisation

- Bird flocks and fish schools show coordinated motion without central control.
- Local interaction can produce highly structured collective behaviour.
- PSO borrows this intuition, but uses it for **optimisation** rather than biological simulation.



Key shift

From coordinated movement in physical space to coordinated search in **solution space**.

Historical intuition

- Reynolds' boids model explained flocking through simple local rules.
- Kennedy and Eberhart then asked whether similar dynamics could solve optimisation problems.
- The central idea: let particles remember good positions and share useful discoveries.

PSO in one sentence

PSO searches by combining **momentum**, **private experience**, and **social learning**.

Optimisation setting

We consider a continuous optimisation problem:

$$\min_{x \in \mathcal{X}} f(x)$$

- The objective may be non-convex or noisy.
- Gradient information may be unavailable or unreliable.
- We therefore need a **black-box search** method.

The intuition of PSO

- Particles start from different candidate solutions.
- Each particle remembers the best point it has found so far.
- The swarm also shares the best discovery made collectively.
- Future motion is biased towards both kinds of memory.

Interpretation

PSO is a search process in which agents learn both from **their own history** and from the **population's history**.

State of a particle

For particle i , we track:

- current position x_i^t
- current velocity v_i^t
- personal best position p_i
- shared best position g (or local best l_i in neighbourhood variants)

Meaning

The particle is not just a point in space; it is a point with **memory** and **directional momentum**.

Fitness

The fitness is the value of the objective function at a particle's current position. For minimisation problems, lower fitness is better; for maximisation problems, higher fitness is better.

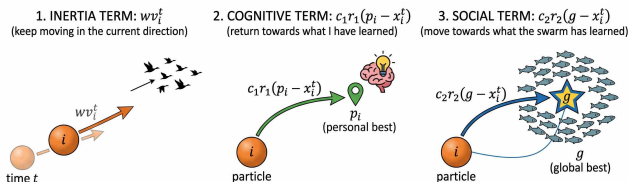
The canonical PSO update

$$v_i^{t+1} = wv_i^t + c_1r_1(p_i - x_i^t) + c_2r_2(g - x_i^t)$$
$$x_i^{t+1} = x_i^t + v_i^{t+1}$$

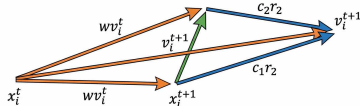
- w : inertia weight
- c_1, c_2 : cognitive and social coefficients
- r_1, r_2 : random variables for stochastic exploration

Interpreting the three terms

- **Inertia term** wv_i^t : keep moving in the current direction.
- **Cognitive term** $c_1r_1(p_i - x_i^t)$: return towards what I have learned.
- **Social term** $c_2r_2(g - x_i^t)$: move towards what the swarm has learned.



THE CANONICAL PSO UPDATE: $v_i^{t+1} = w v_i^t + c_1r_1(p_i - x_i^t) + c_2r_2(g - x_i^t)$



The resultant movement balances personal knowledge, social knowledge, and previous direction to determine the particle's new position
 $x_i^{t+1} = x_i^t + v_i^{t+1}$

Exploration and exploitation

- Large inertia encourages broader exploration.
- Strong social attraction accelerates convergence.
- Strong cognitive attraction preserves individual search trajectories.
- Too much exploitation risks premature convergence.
- Too much exploration slows down progress.

Design tension

PSO works well when it balances **search diversity** with **collective concentration**.

Global-best versus local-best PSO

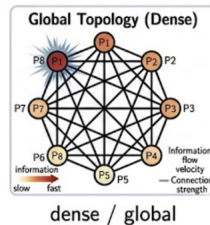
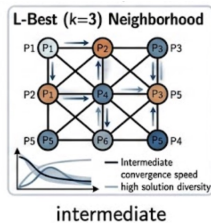
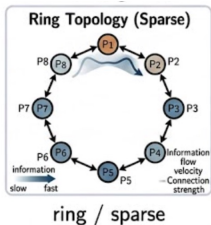
- **Global-best** PSO uses one best position for the whole swarm.
- **Local-best** PSO shares information only within neighbourhoods.
- Local interaction slows information spread but preserves diversity longer.

Trade-off

- global-best: faster convergence
- local-best: better protection against local optima

Population topology matters

- A sparse topology spreads information slowly.
- A dense topology spreads information quickly.
- The neighbourhood graph therefore shapes the search dynamics.



The basic PSO algorithm

1. Initialize particle positions, velocities, and personal bests.
2. Evaluate the objective at each particle and set the initial shared best.
3. **Repeat until a stopping criterion is met:**
 - 3.1 Evaluate the objective at each particle.
 - 3.2 Update each particle's personal best.
 - 3.3 Update the swarm's shared best.
 - 3.4 Update velocities using inertia, self-memory, and social memory.
 - 3.5 Move particles to new positions.

Observation

PSO is a simple iterative loop: particles evaluate, remember, share, and move. The collective search dynamics emerge from repeating these local updates.

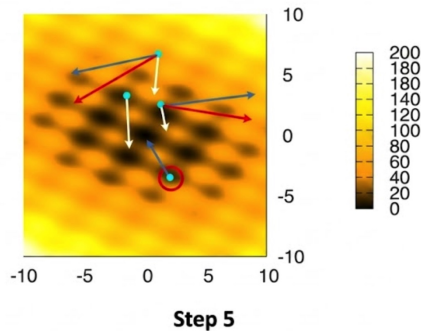
Typical termination criteria

Maximum number of iterations, satisfactory fitness reached, or no significant improvement for several iterations.

PSO Example

Algorithm

1. initialize particles
2. compute fitness
3. update personal best
4. update global best
5. compute new velocities based on momentum, personal best, and global best
6. move particles to new location
7. Go to Step 2 until stopping criteria are met



Strengths of PSO

- derivative-free
- easy to implement
- naturally parallel
- effective on many continuous black-box problems
- conceptually simple but behaviourally rich

Weaknesses of PSO

- may converge too quickly to a suboptimal region
- can be sensitive to hyperparameters
- performance may degrade in very high-dimensional spaces
- lacks the strong guarantees of classical convex optimisation

Takeaway

PSO is powerful because it is simple, but that simplicity also means it relies heavily on good tuning and good search geometry.

Where does PSO sit in the optimisation landscape?

Gradient descent

- uses derivatives
- local update
- strong theory in smooth settings

Evolutionary methods

- population-based
- selection and variation
- broad search family

PSO

- population-based
- social memory
- no gradients required

A unifying summary

One idea to remember

Particle Swarm Optimisation is a distributed search method in which particles move through a continuous space by combining **momentum**, **private memory**, and **shared social information**.

- memory is stored inside particles
- interaction determines convergence behaviour
- topology controls how quickly experience spreads